A Modal Simulation Language for Attribution, Uncertainty, and Conversational Repair

A framework using true/false positives and negatives, modal operators, and explicit repair protocols

Abstract. This paper proposes a compact modal simulation language for analysing conversational claims when a speaker's words, intentions, or emotional meanings are attributed by another person. The framework combines (1) a confusion-matrix model of attribution, (2) modal distinctions between what is actual, possible, necessary, and counterfactual, and (3) a repair protocol that separates observation from inference. Its central rule is that an interpretation must not silently become a fact. False positives and false negatives are treated as different conversational errors, each requiring a different mitigation. A worked example based on the supplied comic illustrates how a paraphrase can be disputed without assuming that either participant is lying.

Keywords: modal logic; speech attribution; uncertainty; false positive; false negative; conversational repair; epistemic humility

1. Problem statement

The comic depicts a short exchange in which one person says, “So I wouldn’t have to be around you.” The other responds, “But I wouldn’t have said that.” The first speaker then says, “That’s not real.” The exchange is useful as a model of a common interpretive problem: a statement about what someone supposedly said or meant is treated as though it were directly observable, even though the original utterance, the paraphrase, and the inferred intention may be different objects.

The framework does not decide who is correct from the comic alone. It instead asks what kind of claim is being made, what evidence supports it, what uncertainty remains, and what response would reduce the chance of compounding an error. The phrase “That’s not real” is especially underspecified: it could reject the alleged quotation, the interpretation, the emotional premise, or the situation as described. A safe protocol asks which proposition is being denied rather than assigning a motive.



Figure 1. The supplied three-panel exchange. The framework treats the dialogue as an example for analysing attribution, not as proof of either participant's private intention.

2. The attribution problem: four classification outcomes

Let p be a proposition about a speaker, such as "Speaker B said the exact sentence X." Let G(p) represent the best available ground-truth status after checking the relevant record, and let A(p) represent whether the interpreter asserts or acts as if p is true. The four outcomes are:

Interpreter's claim	Ground truth: p is true	Ground truth: p is false
Asserts p	True positive (TP) Correctly identifies the statement or meaning.	False positive (FP) Attributes a statement or meaning that is not supported or is false.
Does not assert p	False negative (FN) Misses a statement or meaning that is actually present.	True negative (TN) Correctly refrains from attributing what is absent.

Important qualification. "Truth" must be defined for the specific target. Exact wording, paraphrased content, inferred intention, and perceived emotional impact are four different targets. A paraphrase can be accurate in broad meaning but not be an exact quotation. A person can also deny an intention while acknowledging the words. Therefore, the evaluator must identify the proposition before scoring it.

3. Modal layer: what kind of truth claim is being made?

A confusion matrix evaluates classification against a target. Modal language describes the status or strength of a proposition. The two layers should not be collapsed. This paper uses the following operators:

Operator	Meaning	Example
ACT(p)	p is asserted as actual / observed.	ACT("The message contains these words.")
POSS(p)	p is possible; evidence does not establish it.	POSS("That may have been the intended meaning.")
NEC(p)	p is necessary within the stated model or rules.	NEC("A direct quotation requires a source.")

Operator	Meaning	Example
¬p	The proposition is denied or judged false.	¬ACT(“B said those exact words.”)
CF(p)	p is counterfactual: a hypothetical alternative, not an event report.	CF(“If I had meant that, I would have said it directly.”)
$K_a(p)$	Agent a claims to know p; this is an epistemic claim, not automatic proof.	K_A(“I remember the exact wording.”)
U(p)	The status of p is unresolved / unknown.	U(“Whether that was the intended meaning.”)

A practical language should also mark provenance: OBS(x) for directly observed material, REP(a,p) for a report by agent a, INF(p|e) for an inference from evidence e, and INT(a,p) for a claim about agent a’s intention. This prevents an inference from being silently rewritten as an observation.

4. A compact syntax for modal simulation

The following is a proposed notation, not an established standard. It is designed to be readable in ordinary conversation and precise enough for annotation or simulation.

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CLAIM[agent, target, modality, evidence, confidence, status]
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Example:

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CLAIM[A, “B said X”, ACT, audio_recording, 0.95, VERIFIED]
CLAIM[A, “B intended rejection”, POSS, tone_and_context, 0.45, UNRESOLVED]
CLAIM[B, “I did not say X”, ¬ACT, personal_recollection, 0.70, DISPUTED]
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The confidence values are illustrative degrees of confidence, not calibrated probabilities. A system should not display numerical confidence unless it has a defensible calibration method. “Verified” should be reserved for evidence that actually checks the target proposition.

A useful annotation distinguishes at least these targets:

- LEXICAL: the exact words spoken or written.
- SEMANTIC: the proposition conveyed by those words.
- PRAGMATIC: what the utterance was doing in context (e.g., a request, joke, threat, or boundary).
- INTENTIONAL: what the speaker privately intended.
- IMPACT: how the utterance was received by another person.

These targets can have different truth values. For example, the lexical claim may be false while the semantic paraphrase is broadly accurate. The impact claim may be true even when the inferred intention is false.

5. Confusion-aware mitigation

The goal is not to eliminate all uncertainty. It is to prevent an uncertain attribution from becoming an unquestioned premise, while also avoiding the opposite error of dismissing a real statement or impact. Each error type calls for a distinct response:

Error	Typical conversational form	Mitigation
False positive	“You said / meant X” when X is not established.	Retract or downgrade the claim; quote the source; label inference as inference; invite correction.
False negative	“That never happened” when a statement or impact is evidenced.	Re-check the record; acknowledge the missed evidence; correct the denial without demanding agreement about intent.
True positive	A supported quotation or interpretation.	State the evidence and scope; avoid expanding a verified phrase into an unsupported motive.

Error	Typical conversational form	Mitigation
True negative	No unsupported attribution is made.	Keep the uncertainty visible; do not fill the gap with speculation.

A central asymmetry: false positives can impose an identity, motive, or statement on another person; false negatives can erase a real statement, boundary, or harm. The protocol must guard against both. It should not treat “the person denies it” as proof that it did not happen, nor treat “the person feels certain” as proof that it did.

6. The repair protocol: a sequence of conversational moves

A repair sequence can be represented as a state machine. Each transition should reduce ambiguity or explicitly preserve it.

R0 — Pause: stop escalating the interpretation. Do not repeat the disputed attribution as settled fact.
R1 — Specify: identify the target: exact words, paraphrase, intention, or impact.
R2 — Source: state the evidence source (recording, message, memory, inference, or impression).
R3 — Modalise: mark the claim as ACT, POSS, CF, or U rather than overstating certainty.
R4 — Check: ask the other person whether the wording or interpretation is accurate.
R5 — Repair: if unsupported, retract or revise; if supported, acknowledge the evidence and keep the claim within its scope.
R6 — Boundary: agree on what each person will do next, without requiring agreement about private intention.

The protocol is not a courtroom. It is a way to make conversational claims auditable and revisable. A person may decline to discuss intent; that does not prevent the other person from describing their own experience or setting a boundary.

7. Applying the language to the comic

The exchange can be annotated without deciding which participant is right:

> A: “So I wouldn’t have to be around you.”
> B: “But I wouldn’t have said that.”
> A: “That’s not real.”

Step 1 — Disaggregate the first line. It could be an exact quotation, a paraphrase of an earlier statement, or an inference about what B meant. These are separate claims: LEXICAL(X), SEMANTIC(X), and INTENTIONAL(X).

Step 2 — Interpret B’s denial narrowly. “I wouldn’t have said that” may reject the exact wording, the paraphrase, or the attributed intention. It does not by itself establish which one. The appropriate annotation is U(target) until clarified.

Step 3 — Disambiguate “That’s not real.” The phrase needs an explicit object. Does A mean “that is not what happened,” “that interpretation is inaccurate,” or “that possibility is not credible”? Without an object, the statement risks becoming a global dismissal.

Step 4 — Choose a low-error response. Instead of escalating, either participant can ask for the target and evidence. For example:

> “I may have paraphrased you too strongly. I’m not certain those were your exact words. I was trying to describe how I understood the exchange. Did you mean something different?”

If the original words are documented, a different repair is appropriate:

> “I checked the message, and the wording is there. I’ll keep the claim to what it says; I won’t assume that it proves what you intended.”

If the issue is impact rather than wording:

> “I hear that you disagree with my interpretation. I can still explain how it affected me, without claiming that I know your intention.”

8. Response grammar: a practical phrasebook

Function	Modal response template
Clarify	“Are you denying the exact words, the paraphrase, or the intention I inferred?”
Downgrade certainty	“I may be mistaken; I’m treating that as a possibility, not a fact.”
Request evidence	“What source should we use to check the wording?”
Correct a false positive	“I attributed that too strongly. I withdraw the claim about what you said / meant.”
Correct a false negative	“I checked again. I missed that evidence, and my earlier denial was wrong.”
Separate impact from intent	“I’m describing how I experienced it, not claiming certainty about your motive.”
Set a boundary	“We may disagree about intent. I still need this conversation to remain respectful.”
Close unresolved issue	“We have not established that claim. Let’s leave it unresolved rather than force a conclusion.”

9. Simulation algorithm

A dialogue simulator can use the following decision procedure. It should output a proposed repair, not a verdict about a person's character or hidden mental state.

1. Extract the disputed proposition p and identify its target type.
2. Attach provenance e: direct record, participant report, memory, or inference.
3. Estimate status: supported, contradicted, or unresolved. Do not convert missing evidence into proof of absence.
4. Select the modal operator: ACT for observed facts; POSS for plausible but unconfirmed readings; CF for hypothetical alternatives; U when the target remains unclear.
5. Compare assertion with the best available ground truth to classify TP, FP, FN, or TN.
6. Select repair: clarify, verify, retract, acknowledge, separate impact from intent, or set a boundary.
7. Re-evaluate after new evidence or correction. Preserve an audit trail of what changed and why.

Pseudo-code:

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if target_is_unclear: ask_clarifying_question(); status = U
elif evidence_supports(p): assert_with_scope(p); label = TP
elif evidence_contradicts(p): retract_or_correct(p); label = FP
elif evidence_supports_not(p): refrain_from_asserting(p); label = TN
else: preserve_uncertainty(); status = U

if prior_denial_was_contradicted: acknowledge_missed_evidence(); label = FN
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The final branch is intentionally explicit: a false negative is not corrected by merely becoming more confident. It is corrected by acknowledging the missed evidence and updating the claim.

10. Evaluation criteria and limitations

A simulator can be evaluated on several dimensions: (a) attribution precision, the proportion of asserted attributions that are supported; (b) attribution recall, the proportion of supported attributions that are detected; (c) calibration, whether confidence corresponds to observed accuracy; (d) repair quality, whether corrections are specific and non-escalatory; and (e) scope discipline, whether verified words are kept separate from inferred motives. Precision and recall expose the trade-off between false positives and false negatives. A system that avoids every assertion may reduce false positives but produce many false negatives; a system that asserts every plausible interpretation may do the reverse.

The framework has important limits. Human conversations rarely provide a perfect ground-truth record. Memory can be incomplete, recordings can omit context, and semantic interpretation can remain contested even when exact wording is known. Modal labels do not reveal private intention. Numerical confidence can create false precision. The system must therefore allow “unresolved” as a valid result and must not diagnose, rank, or morally classify participants from a short exchange.

11. Conclusion

A modal simulation language can make conversational interpretation safer by separating what was observed from what was inferred, and by distinguishing actual claims from possible or counterfactual ones. The confusion matrix supplies an error vocabulary: false positives over-attribute; false negatives overlook; true positives identify supported claims; true negatives refrain from unsupported claims. The repair protocol then turns those categories into action: specify the target, identify the evidence, modalise uncertainty, check the interpretation, correct errors, and preserve boundaries. In the comic, the most reliable next move is not to decide who is dishonest. It is to ask what “that” refers to and whether the disagreement concerns words, meaning, intention, or impact.

Appendix A — Minimal modal notation

ACT(p) = p is presented as actual or directly observed.
POSS(p) = p is possible but not established.
NEC(p) = p follows necessarily from stated rules or premises.
¬p = p is denied.
CF(p) = p is counterfactual / hypothetical.
U(p) = p remains unresolved.
OBS(e) = evidence e is directly observed.
REP(a,p) = agent a reports p.
INF(p|e) = p is inferred from evidence e.
INT(a,p) = p is attributed as the intention of agent a.
TP / FP / FN / TN = classification outcomes relative to an explicitly defined target and ground truth.

Appendix B — Design principle

Never let a stronger modality emerge merely because a sentence has been repeated. Repetition does not turn POSS into ACT, an inference into an observation, or a disputed attribution into verified fact. A correction must be allowed to move the system back toward uncertainty.

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